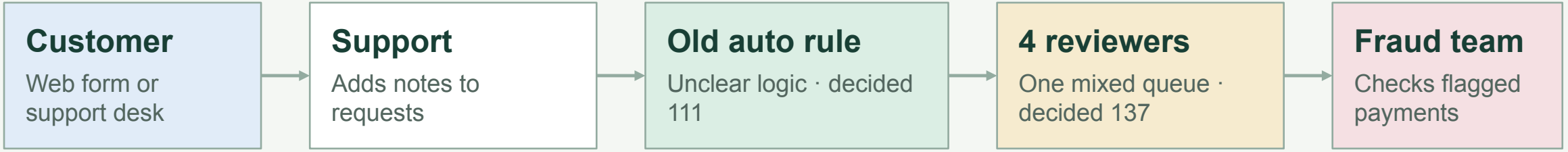
Today: one queue, one forgotten rule

History sample: 248 unique requests from 1 June to 18 August 2026.

WHO IS INVOLVED



The Refund Operations Manager owns the written policy, but nobody quite remembers the auto rule’s logic.

WHERE TIME AND MONEY GO, AND WHERE IT BREAKS

1,119 min

of reviewer time on 137 requests; never-arrived claims took 450 of them

$2,519

8 chargebacks with $25 fees; 7 were staff approvals against evidence already in the notes

7 decisions

inconsistent: 4 change-of-mind refunds after the 14-day limit, 3 rejections overturned on appeal

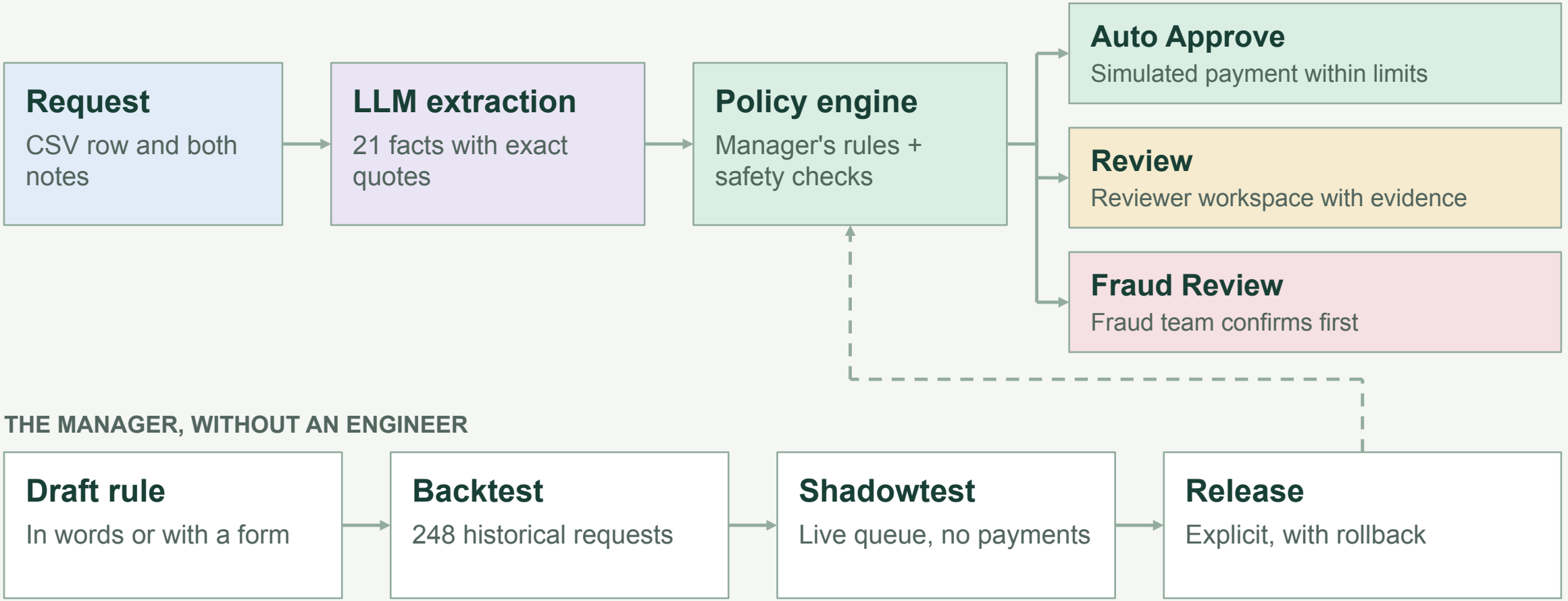
5 payouts

where the old rule ignored notes: 3 damage claims without a photo; today a double refund and a day-22 return

Messy inputs too: duplicates, four amount formats, day-first dates, six fraud-flag spellings and reasons that contradict notes.

After: AI reads, code decides, people judge

Every request gets one explained route. The manager owns the rules and tests every change.



Before: one queue and a rule nobody remembers. After: visible rules, explained outcomes, tested changes.

The line: what the system handles alone

Proposed policy v2. Anything that is not clearly simple goes to a person.

Handled alone

Auto Approve

- Wrong-size apparel up to $100
- Change of mind up to $75, within 14 days
- Apparel or home damage up to $50, or $150 with a photo
- Our mistake, once confirmed: full refund
- Stolen card, once the fraud team confirms: full refund

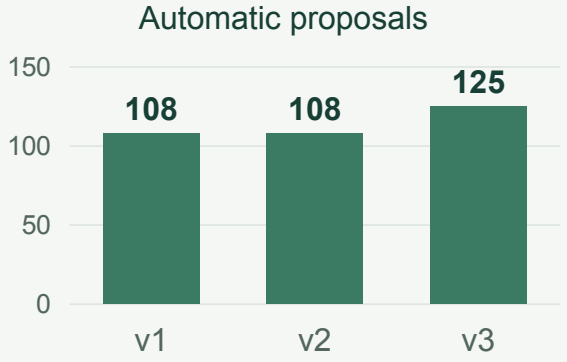
Returns and damage also need no prior refunds, a clear fraud flag and notes that agree.

Goes to a person

Review or Fraud Review

- Anything over $300
- Repeat refunds in the last 12 months
- Delivery proof or a carrier incident
- Electronics, partial or customer-caused damage
- Already refunded, a reason the notes contradict, or embedded instructions
- A fraud flag without confirmation goes to the fraud team

EVIDENCE · 248 REQUESTS

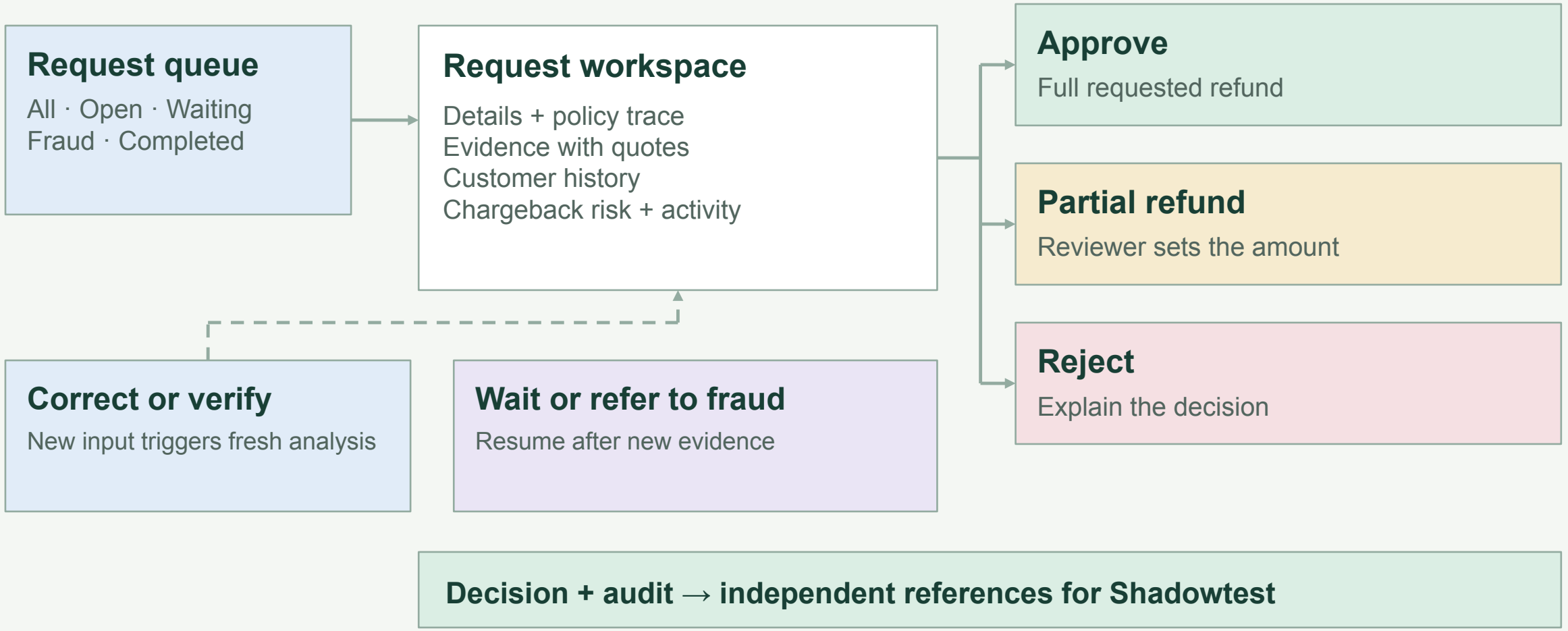


- v1 matches the old rule in 242 of 243 eligible decisions
- v2: same count, different cases: 31 in, 31 out
- v2: $0 above human amounts on 32 assessed, 0 known chargebacks
- v3: $415 above human amounts in 3 cases

214 of 248 outcomes are unknown: the replay supports this line but does not prove it safe.

Reviewer: evidence to decision, in one place

What lands there: everything the policy does not take. Of the 20 pending requests, v2 sends 16 to Review and 2 to Fraud Review.



Manager: change the policy, see the effect first

The brief's two changes, made in a draft version without an engineer.

“From now on, anything over $300 comes to us.”

Amount above $300 → Review · priority 950

Backtest on 248 historical requests: 13 matches. Added to v1 it changes no route: the old rule never auto-approved more than $55.

“From now on, stolen-card cases go straight through.”

Fraud confirmed → Auto Approve · priority 880

Backtest: a rule on reported fraud matches all 11 stolen-card reports, but each still goes to the fraud team first. Confirmation comes later.

Both rules are in v2. AI drafts always return a rule, with vague wording read the safe way. Priority decides between rules: a confirmed stolen card above $300 still comes to Review.

Draft
Edit inline or describe

→

Backtest
248 historical requests

→

Shadowtest
Live queue, no payments

→

Release
Checks pass · rollback ready

AI reads meaning. Code controls money.

The model answers predefined questions with quotes. It never approves, sets an amount or pays.

Source input
Allowed fields + both notes; outcomes excluded

→

LLM extraction
21 predefined facts with exact quotes

→

Validation
Schema + quotes; unknown stays unknown

→

Rules + checks
Python decides route, amount and payment

MODEL, COST AND EVALUATION

gpt-6.1-sol
via OpenAI (default) or OpenRouter; chosen for multilingual notes read against structured fields

≈ $0.005
per request: 268 calls, $1.25 in total (OpenRouter-reported); median 5–7 s

20 / 20
pending acceptance cases; v2 routing matched 20/20 through OpenAI; gpt-5.4-mini passed 14/20

248 / 248
historical extractions complete and validated against quotes

Not AI: routes, amounts, payments, limits, rule discovery and metrics. AI also drafts rules that the manager reviews.

How we would know the line is wrong

Before switching on: Backtest and Shadowtest. After: the same few numbers every week.

THE FEW NUMBERS	GOOD LOOKS LIKE
Automatic share and review minutes	About 44% automatic, as in v2 replay, with reviewer time moving to cases that need judgment
Overpayment on audited automatic refunds	$0 above the human amount; release check: 95% upper error at most 10%
Chargebacks on automatic refunds	Zero, each one investigated; today 8 of 248 requests ended in one
Reviewer overrides and appeals	Falling; today 3 rejections were overturned on appeal

Roll back when

- Any critical violation: a double payment or a limit exceeded
- A chargeback or confirmed overpayment on an automatic refund above tolerance
- Error bound above 10% or stress above $1,000 per 100 approvals
- Automatic share jumps or drops, for example during a carrier incident

Rollback keeps payments and spent limits recorded.

Default release gate: 24 h of Shadowtest, 30 automatic proposals, 30 reviewed customers, 0 critical violations.

Deferred, and what to build next

Deferred from this version; next once there is more independent data.

Better evidence for learning

Collect mature outcomes and independent reviewer labels.
Reserve future time windows and unseen customers for evaluation.

Calibrated risk model

Estimate chargeback probability and dollar loss.
Show uncertainty, calibration and drift.
Keep the manager’s risk tolerances explicit.

More powerful rule discovery

Search richer AND / OR interactions and segments.
Validate future-period support, errors and loss.
Prune overlapping or conflicting candidates.

Proposals stay explainable. Managers inspect the evidence and choose which rules to release.